JT808 Additional Information Item Protocol

JT808 Additional Information Item Protocol

Version History

Version	Date	Modifier	Change Description
1.0.0	2026-04-22	Zhuojie Cai	Initial version
1.0.1	2026-05-07	Zhuojie Cai	Added health data domain (0xF6), including heart rate and blood oxygen (0xF601) composite item; section numbering reorganized
1.0.2	2026-05-11	Zhuojie Cai	Added long data extension domain (0xFA), including tire pressure alarm (0xFA02) composite item
1.0.3	2026-05-15	Zhuojie Cai	Added module extension identifier (0xF401), used to identify the current device connection link type (SOC/MCU)

1.0.4	2026-05-15	Zhuojie Cai	Added oil rod raw data in long data extension domain (0xFA03), used for transparent transmission of oil rod device raw data
1.0.5	2026-05-18	Zhuojie Cai	Added alcohol sensor (0xF506), used for alcohol detection data reporting
1.0.6	2026-05-22	Zhuojie Cai	Added RS485 data extension upload (0xFA04)
1.0.7	2026-05-22	Zhuojie Cai	Added MCU temperature (0x002A), gsensor three-axis value (0x00EA), and video status (0x00F8)

Table of Contents

---

1 Additional Information Item Definition

> **Data Example Description:** Data examples in each additional information item section are divided into two parts:
> **Remark:** For other extension protocols or if you cannot find the required protocol definition, please contact **Software Technical Support Team**.

1.1 Additional Information Item Format (TLV Structure)

Additional information items adopt the TLV (Tag-Length-Value) structure:

Additional Information ID (1 Byte) + Additional Information Length (1 Byte) + Additional Information Data (N Byte)

Field	Data Type	Description and Requirements
Additional Information ID	BYTE	1~255

Additional Information Length	BYTE	—
Additional Information	—	[Nested Unit 1] + ... + [Nested Unit N]

Overall Nested Unit Format:

Level-1 Extension Bit (0xF1) + Data Length (1 Byte) + [Nested Unit 1][Nested Unit 2]...[Nested Unit N]

Each Nested Unit Structure:

Extension Signal Length (1 Byte) + Extension ID (2 Byte) + [Data Group 1][Data Group 2]...[Data Group M]

- Note: Extension signal length = Extension ID (2 Byte) + Data group content

Traversal Rules:

- Traverse nested units in order; each nested unit is self-describing (length + sub-command + content)
- Within each nested unit, traverse multiple data groups in order
- Parse until outer data is exhausted
- **Multiple nested unit types may coexist under the same extension bit** (e.g., 0xF100 and 0xF102 coexisting)

1.2 Additional Information Definition

Business Domain	Level-1 Extension Bit	Sub-ID Range	Available Quantity	Description
Domain 1 (Location and Positioning)	0xF1	0xF100-0xF1FF	256	WiFi, base station, Bluetooth, UWB, etc.
Domain 2 (Power and Energy)	0xF2	0xF200-0xF2FF	256	Voltage, battery level, etc.
Domain 3 (Security and Alarm)	0xF3	0xF300-0xF3FF	256	Extended alarm status bits, etc.

Domain 4 (Device Maintenance)	0xF4	0xF400-0xF4FF	256	IMEI, storage status, version, etc.
Domain 5 (Sensor Data)	0xF5	0xF500-0xF5FF	256	Temperature, humidity, fuel level sensor, load, three-axis acceleration, step count, etc.
Domain 6 (Health Data)	0xF6	0xF600-0xF6FF	256	Heart rate, blood oxygen, etc.
Domain 7 (Long Data Extension)	0xFA	0xFA00-0xFAFF	256	Customer-defined long data

- Continue parsing until bytes read == Length (1 Byte)

1.2.1 Location and Positioning

1.2.1.1 WiFi Information Upload - No SSID (0xF100)

Sub-command: 0xF100

Function Description: WiFi information upload (no SSID, default scenario)

Table 1-1 WiFi Information Upload - No SSID

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	Variable length
1	Extension ID	WORD	F100
3	Data Content	BYTE[N]	See Table 1-1-1

Table 1-1-1 WiFi Information Nested Unit Format

Offset	Field	Data Type	Description and Requirements
0	WiFi MAC Address	BYTE[6]	6-byte MAC address
6	Signal Strength	BYTE	Unit dBm, 0xFF indicates invalid

Item Limit: No more than 10

Note: WiFi MAC address is reported by the device without colons.

HEX Example:F1 0A 09 F1 00 AA BB CC DD EE FF 7F

Parsing: Total additional information length=0A, extension signal length=09, extension ID=F100, data content=AA BB CC DD EE FF (MAC address) 7F (signal strength -127 dBm)

Complete Frame Example (Uplink): 7E 02 00 00 28 09 01 23 38 00 16 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 10 06 04 F1 0A 09 F1 00 A4 C3 F2 00 01 01 BF 5C 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 09 01 23 38 00 16 00 1C 00 01 02 00 00 9E 7E

1.2.1.2 WiFi Information Upload - With SSID (0xF101)

Sub-command: 0xF101

Function Description: WiFi information upload (with SSID)

Table 1-2 WiFi Information Upload - With SSID

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	Variable length
1	Extension ID	WORD	F101
3	Data Content	BYTE[N]	See Table 1-2-1

Table 1-2-1 WiFi Information Nested Unit Format

Offset	Field	Data Type	Description and Requirements
0	WiFi MAC Address	BYTE[6]	6-byte MAC address
6	Signal Strength	BYTE	Unit dBm, 0xFF indicates invalid
7	SSID Length	BYTE	SSID name length, maximum 32 bytes
8	SSID Name	String[N]	SSID name, UTF-8 encoded

Item Limit: No more than 5

HEX Example:F1 10 0F F1 01 AA BB CC DD EE FF 7F 05 48 49 44 45 52

Parsing: Total additional information length=10, extension signal length=0F, extension ID=F101, data content=AA BB CC DD EE FF (MAC address) 7F (signal strength -127 dBm) 05 (SSID length) 4849444552 ("HIDER")

Complete Frame Example (Uplink): 7E 02 00 00 32 09 01 23 38 00 16 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 10 08 55 F1 14 13 F1 01 A4 C3 F2 00 01 01 BF 09 43 4D 43 43 2D 57 69 46 69 27 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 09 01 23 38 00 16 00 30 00 01 02 00 00 B2 7E

1.2.1.3 Base Station Upload Protocol (0xF102)

Sub-command: 0xF102

Function Description: 4G base station data (standardized version, supports internationalization)

Table 1-3 Base Station Upload Protocol

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	Variable length
1	Extension ID	WORD	F102

3	Data Content	BYTE[N]	See Table 1-3-1

Table 1-3-1 Base Station Nested Unit Format

Offset	Field	Data Type	Description and Requirements
0	MCC	WORD	Mobile Country Code, e.g., 460 (China)
2	MNC	WORD	Mobile Network Code, e.g., 00/01/11
4	LAC/TAC	WORD	Location Area Code / Tracking Area Code
6	Cell ID	DWORD	Base station ID, 4 bytes
10	Signal Strength	BYTE	Unit dBm, optional field, 0xFF indicates invalid

Item Limit: Single base station 1 item, multiple base stations no more than 5

HEX Example:F1 0E 0D F1 02 01 C4 00 01 00 00 00 12 34 56 7F

Parsing: Total additional information length=0E, extension signal length=0D, extension ID=F102, data content=01 C4 (MCC=460) 00 01 (MNC=1) 00 00 (LAC=0) 00 00 12 34 56 (Cell ID=305419846) 7F (signal strength -127 dBm)

Complete Frame Example (Uplink): 7E 02 00 00 2B 09 01 23 38 00 16 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 10 08 46 F1 0D 0C F1 02 01 CC 01 03 E8 00 00 30 39 B5 A2 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 09 01 23 38 00 16 00 2F 00 01 02 00 00 AD 7E

1.2.1.4 Bluetooth Device List (0xF103)

Sub-command: 0xF103

Function Description: Bluetooth iBeacon device list (standardized version, for Bluetooth positioning)

Table 1-4 Bluetooth Device List

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	Variable length
1	Extension ID	WORD	F103
3	Data Content	BYTE[N]	See Table 1-4-1

Table 1-4-1 Bluetooth Nested Unit Format

Offset	Field	Data Type	Description and Requirements
0	Bluetooth MAC Address	BYTE[6]	6-byte MAC address
6	Battery Level	BYTE	Unit 1%, range 0-100, 0xFF indicates invalid
7	Major	WORD	iBeacon Major value, network byte order
9	Minor	WORD	iBeacon Minor value, network byte order
11	Signal Strength	BYTE	Unit dBm, 0xFF indicates invalid

Item Limit: No more than 5

Note: Bluetooth MAC address is reported by the device without colons.

HEX Example:F1 0F 0E F1 03 AA BB CC DD EE FF 64 00 01 00 02 7F

Parsing: Total additional information length=0F, extension signal length=0E, extension ID=F103, data content=AA BB CC DD EE FF (Bluetooth MAC address) 64 (battery level

100%) 00 01 (Major=1) 00 02 (Minor=2) 7F (signal strength -127 dBm)

Complete Frame Example (Uplink): 7E 02 00 00 2D 09 01 23 38 00 16 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 10 06 32 F1 0F 0E F1 03 00 00 00 00 00 00 50 00 64 00 C8 BF 07 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 09 01 23 38 00 16 00 20 00 01 02 00 00 A2 7E

1.2.1.5 UWB Coordinate Precision Area (0xF104)

Sub-command: 0xF104

Function Description: UWB coordinate precision area (standardized version)

Table 1-5 UWB Coordinate Precision Area

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	Variable length
1	Extension ID	WORD	F104
3	Data Content	BYTE[N]	See Table 1-5-1

Table 1-5-1 UWB Nested Unit Format

Offset	Field	Data Type	Description and Requirements
0	Coordinate Type	BYTE	0: Unknown 1: 2D 2: 3D
1	X Coordinate	DWORD	Unit mm, signed integer, 0xFFFFFFFF indicates invalid
5	Y Coordinate	DWORD	Unit mm, signed integer, 0xFFFFFFFF indicates invalid

9	Z Coordinate	DWORD	Unit mm, signed integer (for 3D), 0xFFFFFFFF indicates invalid
13	Precision FOM	BYTE	Unit mm, 0xFF indicates invalid
14	Area Number	WORD	Area identifier

Data Length: 16 bytes (Coordinate Type 1 + X 4 + Y 4 + Z 4 + FOM 1 + Area Number 2 = 16 bytes, Z coordinate field always exists but only meaningful for 3D)

HEX Example:F1 10 0F F1 04 00 00 12 34 00 00 56 78 00 00 9A BC 05 00 01

Parsing: Total additional information length=10, extension signal length=0F, extension ID=F104, data content=00 (coordinate type=2D) 00 00 12 34 (X=4660mm) 00 00 56 78 (Y=22136mm) 00 00 9A BC (Z invalid) 05 (FOM=5mm) 00 01 (area number=1)

Complete Frame Example (Uplink): 7E 02 00 00 31 09 01 23 38 00 16 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 10 06 16 F1 13 12 F1 04 02 00 00 03 E8 00 00 07 D0 00 00 01 2C 32 00 01 5B 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 09 01 23 38 00 16 00 1D 00 01 02 00 00 9F 7E

1.2.2 Power and Energy

1.2.2.1 Battery Level (0xF200)

Sub-command: 0xF200

Function Description: Battery level (standardized version)

Table 1-6 Battery Level

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	4
1	Extension ID	WORD	F200
3	Battery Level	BYTE	Unit 1%, range 0-100, 0xFF indicates invalid

4	Charging Status	BYTE	Bit definition see Table 1-6-1

Table 1-6-1 Charging Status Bit Definition

Bit	Field	Description
bit0	Charging Flag	0: Not charging; 1: Charging
bit1	Connection Status	0: Not connected to power; 1: Connected to power
bit2	Full Status	0: Not full; 1: Full
bit3~bit7	Reserved	—

HEX Example:F2 05 04 F2 00 64 07

Parsing: Total additional information length=05, extension signal length=04, extension ID=F200, data content=64 (battery level 100%) 07 (charging status: bit0=1 charging, bit1=1 power connected, bit2=0 not full)

Complete Frame Example (Uplink): 7E 02 00 00 23 09 01 23 38 00 16 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 10 06 43 F2 05 04 F2 00 50 00 68 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 09 01 23 38 00 16 00 22 00 01 02 00 00 A0 7E

1.2.2.2 Voltage (0xF201)

Sub-command: 0xF201

Function Description: Voltage information (standardized version, integrates power supply voltage, battery voltage, ADC voltage)

Table 1-7 Voltage

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	Variable length

1	Extension ID	WORD	F201
3	Data Content	BYTE[N]	See Table 1-7-1

Table 1-7-1 Voltage Nested Unit Format

Offset	Field	Data Type	Description and Requirements
0	Voltage Type	BYTE	Bit definition see Table 1-7-2
1	ADC Channel Number	BYTE	When voltage type is ADC channel, indicates channel number (1~4); fill 0 for other types
2	Voltage Value	WORD	Unit mV

Table 1-7-2 Voltage Type Bit Definition

Bit	Field	Description
bit0~bit1	Voltage Type	00: Power supply voltage; 01: Battery voltage; 10: ADC channel
bit2~bit7	Reserved	—

Note: Multi-channel voltage acquisition is supported through repeated nested units.

HEX Example:F2 07 06 F2 01 01 0C D4

Parsing: Total additional information length=07, extension signal length=06, extension ID=F201, data content=01 (voltage type=power supply voltage) 01 (ADC channel number=1) 0C D4 (voltage value=3300 mV)

Complete Frame Example (Uplink): 7E 02 00 00 25 09 01 23 38 00 16 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 10 07 00 F2 07 06 F2 01 00 01 30 70 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 09 01 23 38 00 16 00 25 00 01 02 00 00 A7 7E

1.2.3 Security and Alarm

1.2.3.1 Extended Alarm Status Bits (0xF300)

Sub-command: 0xF300

Function Description: Extended alarm status bits (standardized version)

Table 1-8 Extended Alarm Status Bits

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	6
1	Extension ID	WORD	F300
3	Alarm Status	DWORD	State[31~0], bit definition see Table 1-8-1

Table 1-8-1 Alarm Status Bit Definition

Bit	Field	Description
bit0	batterySwitch	1: Battery switch-off; 0: Battery switch-on
bit1	Sleep Status	1: Terminal normal status; 0: Terminal sleep status
bit2	Card Swipe Status	1: Normal; 0: Driving without card swipe
bit3	Overspeed Status	1: Normal; 0: Overspeed during time period
bit4	collideAlarm	1: Normal; 0: Collision alarm
bit5	Evaluator Status	1: Evaluator connected normal; 0: Evaluator connection abnormal

bit6	Top Cover Status	1: Top cover closed; 0: Top cover not closed
bit7	Compartment Status	1: Compartment not lifted; 0: Compartment lifted
bit8	accelerateAlarm	1: Normal; 0: Harsh acceleration alarm
bit9	slowDownAlarm	1: Normal; 0: Harsh braking alarm
bit10	Acceleration Sensor Status	1: Acceleration sensor normal; 0: Acceleration sensor abnormal
bit11	illegalMove	1: Normal; 0: Illegal displacement alarm
bit12	Removal Status	1: Normal; 0: Illegal removal
bit13	High Temperature Alarm	1: Normal; 0: High temperature alarm
bit14	Low Temperature Alarm	1: Normal; 0: Low temperature alarm
bit15	shockAlarm	1: Normal; 0: Vibration alarm
bit16	Cut Alarm	1: Normal; 0: Cut alarm
bit17	Power Loss Alarm	1: Normal; 0: Power loss alarm
bit18	Idle Alarm	1: Normal; 0: Idle alarm
bit19	LED Display Screen Status	1: LED display screen connected normal; 0: LED display screen fault
bit20	Taximeter Status	1: Taximeter connected normal; 0: Taximeter fault

bit21	Multifunction Screen Status	1: Multifunction screen connected normal; 0: Multifunction screen fault
bit22	Material Shortage Alarm	1: Normal; 0: Material shortage alarm
bit23	Video Connection Status	1: Normal; 0: Abnormal
bit24	ACC Status	1: ACC normal; 0: ACC disconnection alarm
bit25	swerveAlarm	1: Normal; 0: Harsh steering alarm
bit26	ACC Switch Alarm	1: Normal; 0: ACC transition from off to on alarm
bit27	Main Power Switch Status	1: Main power switch normal; 0: Main power switch abnormal
bit28	Burner Switch Status	1: Burner switch normal; 0: Burner switch abnormal
bit29	Vibration Alarm Switch Status	1: Vibration alarm switch off; 0: Vibration alarm switch on
bit30	pseudoLbsState	1: Normal; 0: Pseudo-base station detected
bit31	pseudoLbs	Pseudo-base station alarm

HEX Example:F3 07 06 F3 00 00 00 00 00

Parsing: Total additional information length=07, extension signal length=06, extension ID=F300, data content=00 00 00 00 (all alarm status bits are 0, indicating normal with no alarms)

Complete Frame Example (Uplink): 7E 02 00 00 25 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 14 29 29 F3 07 06 F3 00 FF FF FF FE F1 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 01 00 01 02 00 00 0C 7E

1.2.3.2 Extended Alarm Data Bits 2 (0xF301)

Sub-command: 0xF301

Function Description: Extended alarm data bits 2 (standardized version, integrates original 0x00C5)

Table 1-9 Extended Alarm Data Bits 2

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	6
1	Extension ID	WORD	F301
3	Alarm Status	DWORD	State[0~31], bit definition see Table 1-9-1

Table 1-9-1 Alarm Status Bit Definition

Bit	Field	Description
Bit0	softSwitch	1: Software switch-off; 0: Software switch-on
Bit1	wayBillModel	1: Waybill mode-off; 0: Waybill mode-on
Bit2	isCharge	1: Not charging; 0: Charging
Bit3~Bit4	posStateOne	Positioning status Bit3=0 Bit4=0: Not positioned (neither GPS nor WIFI positioning) Bit3=1 Bit4=0: GPS positioning Bit3=0 Bit4=1: WIFI positioning

Bit5	Unloading Status	1: Not unloaded; 0: Unloaded
Bit6	shockAlarm	1: Normal; 0: Vibration alarm
Bit7	collideAlarm	1: Normal; 0: Collision alarm
Bit8	Arming Status	1: Disarmed; 0: Armed
Bit9	Motion Status	1: Stationary; 0: Moving
Bit10~Bit11	Work Mode	Bit10=1 Bit11=1: Normal work mode Bit10=0 Bit11=1: Deep power saving mode Bit10=1 Bit11=0: Smart power saving mode
Bit12	Force GPS On	1: Not force GPS on; 0: Force GPS on
Bit13	Wired/Wireless Mode	1: Wired mode; 0: Wireless mode
Bit14	Light Sensor Status	0: Light detected; 1: No light detected
Bit15	G-Sensor Mode	1: G-Sensor mode; 0: Non G-Sensor mode
Bit16	Infrared Status	1: Infrared normal; 0: Infrared abnormal
Bit17	Pressure Status	1: Pressure normal; 0: High pressure alarm
Bit18	Tachograph Status	1: Tachograph normal; 0: Tachograph abnormal

Bit19~Bit20	Positioning Mode	Bit19=0 Bit20=1: Single BeiDou positioning Bit19=1 Bit20=0: Single GPS positioning Bit19=0 Bit20=0: Dual-mode positioning
Bit21	Fuel Circuit Function	0: Overspeed warning; 1: Fuel circuit function
Bit22	Fuel Circuit Status	0: Fuel circuit disconnected; 1: Fuel circuit normal
Bit23	Lane Restriction Alarm	0: Lane restriction alarm; 1: Normal
Bit24	Undervoltage Protection Status	0: Undervoltage protection status; 1: Normal status
Bit25	Low Power Mode	0: Low power mode; 1: Non low power mode
Bit26	Customer/ Mode	0: Customer mode; 1: mode
Bit27	Left Turn Overspeed Alarm	0: Left turn overspeed alarm; 1: Normal
Bit28	Right Turn Overspeed Alarm	0: Right turn overspeed alarm; 1: Normal
Bit29	Infrared 1 Trigger	0: Infrared 1 triggered; 1: Infrared 1 not triggered
Bit30	Infrared 2 Trigger	0: Infrared 2 triggered; 1: Infrared 2 not triggered
Bit31	Infrared 3 Trigger	0: Infrared 3 triggered; 1: Infrared 3 not triggered

HEX Example:F3 07 06 F3 01 FF FF FF FF

Parsing: Total additional information length=07, extension signal length=06, extension ID=F301, data content=FF FF FF FF (all status bits are 1, indicating all alarm/abnormal states)

Complete Frame Example (Uplink): 7E 02 00 00 25 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 14 29 43 F3 07 06 F3 01 FF FF FF FE 9A 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 02 00 01 02 00 00 0F 7E

1.2.3.3 Door Sensor Alarm Status (0xF302)

Sub-command: 0xF302

Function Description: Door sensor alarm status (standardized version, integrates original 0x013D)

Table 1-10 Door Sensor Alarm Status

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	Variable length
1	Extension ID	WORD	F302
3	Data Content	BYTE[N]	See Table 1-10-1

Table 1-10-1 Door Sensor Nested Unit Format

Offset	Field	Data Type	Description and Requirements
0	Device Status	BYTE	Bit definition see Table 1-10-2
1	Voltage	WORD	Unit mV
3	MAC ID	BYTE[6]	MAC address

Table 1-10-2 Device Status Bit Definition

Bit	Field	Description
bit0	Door Sensor Status	0: Closed; 1: Open

bit1	Enabled Status	1: Enabled; 0: Disabled
bit2	Connection Status	1: Present; 0: Lost
bit3~bit7	Reserved	—

Channel Description: Multi-channel supported, achieved through multiple nested units

Note: Door sensor status changes can trigger corresponding alarm flags.

HEX Example:F3 0C 0B F3 02 01 0C D6 AA BB CC DD EE FF

Parsing: Total additional information length=0C, extension signal length=0B, extension ID=F302, data content=01 (device status: bit0=1 door open) 0C D6 (voltage=3030 mV) AA BB CC DD EE FF (MAC address)

Complete Frame Example (Uplink): 7E 02 00 00 2A 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 14 30 13 F3 0C 0B F3 02 07 0B B8 00 00 00 00 00 00 6C 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 03 00 01 02 00 00 0E 7E

1.2.4 Device Maintenance

1.2.4.1 IMEI (0xF400)

Sub-command: 0xF400

Function Description: Device IMEI number

Table 1-11 IMEI

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	17
1	Extension ID	WORD	F400
3	IMEI	BYTE[15]	15-byte ASCII code, device IMEI number

HEX Example:F4 12 11 F4 00 38 36 35 37 33 30 30 37 36 31 31 36 36 38 36

Parsing: Total additional information length=12, extension signal length=11, extension ID=F400, data content=38 36 35 37 33 30 30 37 36 31 31 36 36 38 36 (ASCII: "865730076116686")

Complete Frame Example (Uplink): 7E 02 00 00 30 12 34 56 78 90 12 00 01 00 00 00 00 00 00 00 01 01 65 EC 15 06 BE 20 80 00 00 02 58 00 5A 26 05 25 10 48 30 F4 12 11 F4 00 31 32 33 34 35 36 37 38 39 30 31 32 33 34 35 60 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 02 00 01 02 00 00 0F 7E

1.2.4.2 Module Extension Identifier (0xF401)

Sub-command: 0xF401

Function Description: Module extension identifier (used to identify the current device connection link type)

Table 1-12 Module Extension Identifier

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	3
1	Extension ID	WORD	F401
3	Link Type	BYTE	0: SOC link 1: MCU link

HEX Example:F4 04 03 F4 01 01

Parsing: Total additional information length=04, extension signal length=03, extension ID=F401, data content=01 (link type=1, MCU link)

Complete Frame Example (Uplink): 7E 02 00 00 22 12 34 56 78 90 12 00 01 00 00 00 00 00 00 00 01 01 65 EC 15 06 BE 20 80 00 00 02 58 00 5A 26 05 25 11 06 08 F4 04 03 F4 01 01 31 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 02 00 01 02 00 00 0F 7E

1.2.5 Sensor Data

1.2.5.1 Fuel Level Sensor (0xF500)

Sub-command: 0xF500

Function Description: Fuel level sensor (standardized version)

Table 1-12 Fuel Level Sensor

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	Variable length
1	Extension ID	WORD	F500
3	Data Content	BYTE[N]	See Table 1-12-1

Table 1-12-1 Fuel Level Nested Unit Format

Offset	Field	Data Type	Description and Requirements
0	Channel ID	BYTE	Fuel tank channel number, starting from 1, 0xFF indicates invalid
1	Fuel Percentage	BYTE	Unit 1%, range 0-100, 0xFF indicates invalid
2	Fuel Volume in Liters	DWORD	Unit 0.01L, **Actual value = Protocol value / 100 (L)**, 0xFFFFFFFF indicates invalid

Item Limit: Up to 6 fuel tanks

Note: The device is responsible for completing the liquid level to fuel volume conversion based on fuel tank parameters, outputting physical quantities directly.

HEX Example:F5 09 08 F5 00 01 64 00 00 27 10

Parsing: Total additional information length=09, extension signal length=08, extension ID=F500, data content=01 (channel ID=1) 64 (fuel percentage=100%) 00 00 27 10 (fuel volume in liters=10000 = 100.00L)

Complete Frame Example (Uplink): 7E 02 00 00 27 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 14 31 52 F5 09 08 F5 00 01 32 00 00 13 88 39 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 05 00 01 02 00 00 08 7E

1.2.5.2 Load Sensor (0xF501)

Sub-command: 0xF501

Function Description: Load sensor (standardized version)

Table 1-13 Load Sensor

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	Variable length
1	Extension ID	WORD	F501
3	Data Content	BYTE[N]	See Table 1-13-1

Table 1-13-1 Load Nested Unit Format

Offset	Field	Data Type	Description and Requirements
0	External Device ID	WORD	RS485 address, fill 0 if this interface does not exist
2	Channel ID	BYTE	Sensor channel number, starting from 1, 0xFF indicates invalid
3	Sensor Status	BYTE	Bit definition see Table 1-13-2

4	Load Value	DWORD	Unit 0.1kg, **Actual value = Protocol value / 10 (kg)**, 0xFFFFFFFF indicates invalid
8	Load Status	BYTE	Bit definition see Table 1-13-3

Table 1-13-2 Sensor Status Bit Definition

Bit	Field	Description
bit0	Connection Status	0: Abnormal; 1: Normal
bit1	Sensor Fault	0: Normal; 1: Fault
bit2	Data Valid	0: Invalid; 1: Valid
bit3~bit7	Reserved	—

Table 1-13-3 Load Status Bit Definition

Bit	Field	Description
bit0~bit2	Load Status	000: Unknown; 001: Empty; 010: Light load; 011: Normal (standard load); 100: Full load; 101: Overload; 110: Unloading; 111: Loading
bit3~bit7	Reserved	—

Item Limit: Up to 8 sensors

Note: Load-related signals uniformly use this structure.

HEX Example:F5 0C 0B F5 01 00 01 07 00 00 03 E8 00 00

Parsing: Total additional information length=0C, extension signal length=0B, extension ID=F501, data content=01 00 (external device ID=256) 01 (channel ID=1) 07 (sensor status: connection normal) 00 00 03 E8 (load value=1000 = 100.0 kg) 00 00 (load status=empty)

Complete Frame Example (Uplink): 7E 02 00 00 2A 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 14 36 22 F5 0C 0B F5 01 00 01 01 05 00 00 03 E8 02 00 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 06 00 01 02 00 00 0B 7E

1.2.5.3 Temperature Sensor (0xF502)

Sub-command: 0xF502

Function Description: Temperature sensor (standardized version)

Table 1-14 Temperature Sensor

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	Variable length
1	Extension ID	WORD	F502
3	Data Content	BYTE[N]	See Table 1-14-1

Table 1-14-1 Temperature Nested Unit Format

Offset	Field	Data Type	Description and Requirements
0	Channel ID	BYTE	Channel number, starting from 1, 0xFF indicates invalid
1	Temperature	WORD	int16_t two's complement, unit 0.01°C, **Actual value = Protocol value / 100 (°C)**, 0x8000 indicates invalid

Channel Description: Multi-channel supported, achieved through multiple nested units

Note: Temperature-related signals uniformly use this structure.

HEX Example:F5 06 05 F5 02 02 27 10

Parsing: Total additional information length=06, extension signal length=05, extension ID=F502, data content=02 (channel ID=2) 27 10 (temperature value=10000 = 25.60°C, int16 two's complement)

Complete Frame Example (Uplink): 7E 02 00 00 26 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 14 37 01 F5 08 07 F5 02 01 00 19 00 FF 2E 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 07 00 01 02 00 00 0A 7E

1.2.5.4 Humidity Sensor (0xF503)

Sub-command: 0xF503

Function Description: Humidity sensor (standardized version)

Table 1-15 Humidity Sensor

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	Variable length
1	Extension ID	WORD	F503
3	Data Content	BYTE[N]	See Table 1-15-1

Table 1-15-1 Humidity Nested Unit Format

Offset	Field	Data Type	Description and Requirements
0	Channel ID	BYTE	Channel number, starting from 1, 0xFF indicates invalid

1	Humidity	WORD	Unit 0.01%, **Actual value = Protocol value / 100 (%)**, 0xFF00 indicates this channel is offline

Channel Description: Multi-channel supported (up to 16 channels)

Note: Humidity-related signals uniformly use this structure.

HEX Example:F5 06 05 F5 03 01 27 10

Parsing: Total additional information length=06, extension signal length=05, extension ID=F503, data content=01 (channel ID=1) 27 10 (humidity value=10000 = 100.00%)

Complete Frame Example (Uplink): 7E 02 00 00 26 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 14 37 55 F5 08 07 F5 03 01 17 0C 00 00 86 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 08 00 01 02 00 00 05 7E

1.2.5.5 Step Count Accumulation (0xF504)

Sub-command: 0xF504

Function Description: Step count accumulation (standardized version)

Table 1-16 Step Count Accumulation

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	6
1	Extension ID	WORD	F504
3	Step Count	DWORD	Unit 1 step, unsigned integer, 0xFFFFFFFF indicates invalid

Note: Step count-related signals uniformly use this structure.

HEX Example:F5 07 06 F5 04 00 01 86 A0

Parsing: Total additional information length=07, extension signal length=06, extension ID=F504, data content=00 01 86 A0 (step count=100000 steps)

Complete Frame Example (Uplink): 7E 02 00 00 25 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 14 45 14 F5 07 06 F5 04 00 00 27 10 92 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 0B 00 01 02 00 00 06 7E

1.2.5.6 Barometric Pressure Sensor (0xF505)

Sub-command: 0xF505

Function Description: Barometric pressure sensor

Table 1-17 Barometric Pressure Sensor

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	6
1	Extension ID	WORD	F505
3	Barometric Pressure Value	DWORD	Unit Pa (Pascal), actual value = protocol value, 0xFFFFFFFF indicates invalid

Note: 1 hPa = 100 Pa. Standard atmospheric pressure is 101325 Pa (protocol value 101325)

HEX Example:F5 07 06 F5 05 00 01 89 BC

Parsing: Total additional information length=07, extension signal length=06, extension ID=F505, data content=00 01 89 BC (barometric pressure value=101325 Pa, standard atmospheric pressure)

Complete Frame Example (Uplink): 7E 02 00 00 27 12 34 56 78 90 12 00 01 00 00 00 00 00 00 00 01 01 65 EC 15 06 BE 20 80 00 00 02 58 00 5A 26 05 25 11 08 38 F5 09 08 F5 05 00 00 00 00 00 00 97 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 0B 00 01 02 00 00 06 7E

1.2.5.7 Alcohol Sensor (0xF506)

Sub-command: 0xF506

Function Description: Alcohol sensor

Data Format:

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	N, depends on the actual length of the alcohol concentration string
1	Extension ID	WORD	F506
3	Alcohol Concentration String	ASCII string	Alcohol concentration string assembled by the device, e.g., "200mg/100ml"

Note: The device completes the unit conversion and assembles it into a string for reporting. The platform is only responsible for storage and display, without numerical parsing.

HEX Example:F5 0E 0D F5 06 32 30 30 6D 67 2F 31 30 30 6D 6C

Parsing: Total additional information length=0E, extension signal length=0D, extension ID=F506, data content=32 30 30 6D 67 2F 31 30 30 6D 6C (ASCII: "200mg/100ml", 11 bytes)

1.2.6 Health Data

1.2.6.1 Heart Rate and Blood Oxygen (0xF601)

Sub-command: 0xF601

Function Description: Heart rate and blood oxygen saturation

Table 1-18 Heart Rate and Blood Oxygen

Offset	Field	Data Type	Description

0	Extension Signal Length	BYTE	4
1	Extension ID	WORD	F601
3	Heart Rate Value	BYTE	Unit bpm, range 0-255, 0xFF indicates invalid
4	Blood Oxygen Value	BYTE	Unit %, range 0-100, 0xFF indicates invalid

HEX Example:F6 05 04 F6 01 64 62

Parsing: Total additional information length=05, extension signal length=04, extension ID=F601, data content=64 (heart rate=100 bpm) 62 (blood oxygen=98%)

Complete Frame Example (Uplink): 7E 02 00 00 27 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 14 47 51 F5 09 08 F5 05 00 00 00 00 00 00 E1 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 0C 00 01 02 00 00 01 7E

1.2.7 Long Data Extension

1.2.7.1 GSensor Three-Axis Value (0xFA01)

Sub-command: 0xFA01

Function Description: GSensor three-axis value report (collision event data)

Data Format:

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	Extension ID and data content length, variable length
2	Extension ID	WORD	FA01

4	Event Type	BYTE	1 Byte
5	Three-Axis Data	BYTE[3]×40	X Y Z three-axis values, 3 bytes per group, 40 groups total

Data Length Calculation:

- Data content = 1 (event type) + 40 groups × 3 bytes = 121 Byte
- Extension signal length = 2 (extension ID) + 121 = 123 Byte

HEX Example:FA 7C 7B FA 01 01 BA BB DB ... BA BB DB

Parsing: Total additional information length=7C (124), extension signal length=7B (123), extension ID=FA01, event=01, three-axis data=BA BB DB... (40 groups total)

Complete Frame Example (Uplink): 7E 02 00 00 9B 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 14 48 36 FA 7D 01 7C 7B FA 01 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 4B 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 0D 00 01 02 00 00 00 7E

1.2.7.2 Tire Pressure Alarm (0xFA02)

Sub-command: 0xFA02

Function Description: Tire pressure sensor data report (alarm event data)

Data Format:

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	Variable length
1	Extension ID	WORD	FA02
3	Total Tire Count	BYTE	Sensor count, range 1-22

4	Tire 1 Data	BYTE[4]	See Table 1-19-1
8	Tire 2 Data	BYTE[4]	See Table 1-19-1
...	...	...	...
4+(N-1)×4	Tire N Data	BYTE[4]	See Table 1-19-1

Data Length Calculation:

- Data content = 1 (total tire count) + N×4 Byte
- Extension signal length = 2 (extension ID) + 1 + N×4 Byte

Table 1-19-1 Tire Data Format

Offset	Field	Data Type	Description
0	Tire Position	BYTE	01~22 (Tire 1 - Tire 22), sorted from left to right, front to back
1	Pressure Value	BYTE	**Actual pressure value P = P0 × 5.5 - 100 (KPA)**, P0 is the uploaded raw value
2	Temperature Value	BYTE	Actual temperature T(°C) = T0 - 50, T0 is the uploaded raw value (0 ~ 255). **When device uploads 0, it means -50°C.** Converted temperature range: -50~205°C
3	Status	BYTE	Bit definition see Table 1-19-2

Table 1-19-2 Status Bit Definition

Bit	Field	Description
bit0	High Temperature Alarm	0: Normal; 1: High temperature alarm
bit1	Low Pressure Alarm	0: Normal; 1: Low pressure alarm
bit2	High Pressure Alarm	0: Normal; 1: High pressure alarm
bit3	Air Leakage Alarm	0: Normal; 1: Air leakage alarm
bit4	Low Battery Voltage Alarm	0: Normal; 1: Low battery voltage
bit5	Signal Exception Alarm	0: Normal; 1: Signal loss alarm
bit6~bit7	Reserved	—

HEX Example:FA 0C 0B FA 02 02 01 2C 28 11 02 30 2A 10

Parsing: Total additional information length=0C (12), extension signal length=0B (11), extension ID=FA02, total tire count=02 (2 tires)

Tire 1 data: 01 (position=Tire 1) 2C (pressure value P0=44, actual P=44×5.5-100=142KPA) 28 (temperature value T0=40, actual T=40-50=-10°C) 11 (status: bit0=1 high temperature alarm, bit4=1 low battery voltage)

Tire 2 data: 02 (position=Tire 2) 30 (pressure value P0=48, actual P=48×5.5-100=164KPA) 2A (temperature value T0=42, actual T=42-50=-8°C) 10 (status: bit4=1 low battery voltage)

Complete Frame Example (Uplink): 7E 02 00 00 2B 12 34 56 78 90 12 00 01 00 00 00 00 00 00 00 01 01 65 EC 15 06 BE 20 80 00 00 02 58 00 5A 26 05 25 11 38 52 FA 0D 0C 0B FA 02 02 01 2C 28 0B 02 30 2A 0A 4D 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 03 00 01 02 00 00 0E 7E

1.2.7.3 Oil Rod Raw Data (0xFA03)

Sub-command: 0xFA03

Function Description: Oil rod raw data report

Data Format:

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	Variable length
1	Extension ID	WORD	FA03
3	Data Content	BYTE[N]	RS232 transparent transmission data, N≤252

Data Length Calculation:

- Extension signal length = 2 (extension ID) + N (N is data content length, N≤252)

HEX Example:FA 0B 0A FA 03 01 02 03 04 05 06 07 08

Parsing: Total additional information length=0B (11), extension signal length=0A (10), extension ID=FA03, data content=01 02 03 04 05 06 07 08 (8-byte transparent transmission data)

Complete Frame Example (Uplink): 7E 02 00 00 2A 12 34 56 78 90 12 00 01 00 00 00 00 00 00 00 01 01 65 EC 15 06 BE 20 80 00 00 02 58 00 5A 26 05 25 11 52 10 FA 0C 0B 0A FA 03 01 02 03 04 05 06 07 08 74 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 02 00 01 02 00 00 0F 7E

1.2.7.4 RS485 Data Extension Upload (0xFA04)

Sub-command: 0xFA04

Function Description: RS485 data extension upload (transparent transmission)

Data Format:

Offset	Field	Data Type	Description
0	Extension Signal Length	BYTE	Variable length
1	Extension ID	WORD	FA04

3	Data Content	BYTE[N]	RS485 transparent transmission data, N≤252

Data Length Calculation:

- Extension signal length = 2 (extension ID) + N (N is data content length, N≤252)

HEX Example:FA 0B 0A FA 04 01 02 03 04 05 06 07 08

Parsing: Total additional information length=0B (11), extension signal length=0A (10), extension ID=FA04, data content=01 02 03 04 05 06 07 08 (8-byte transparent transmission data)

Complete Frame Example (Uplink): 7E 02 00 00 2A 12 34 56 78 90 12 00 01 00 00 00 00 00 00 00 01 01 65 EC 15 06 BE 20 80 00 00 02 58 00 5A 26 05 25 11 55 12 FA 0C 0B 0A FA 04 01 02 03 04 05 06 07 08 76 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 02 00 01 02 00 00 0F 7E

Appendix A 0xEB Extension Protocol Overview

A.1 0xEB Extension Protocol Overview

A.1.1 Unified Envelope Format

0xEB extension signals adopt a unified envelope format:

Extension Length (2 Byte) + Extension ID (2 Byte) + Data Content (N Byte)

A.1.2 Differences from 0xF1-F5 Extension Bits

Item	0xEB Extension Bit	0xF1-F5 Extension Bits
Extension Signal Length	2 Byte	1 Byte
Level-1 Extension Bit	0xEB	0xF1/0xF2/0xF3/0xF4/0xF5
Applicable Scenario	For historical compatibility	Recommended for new devices

A.1.3 Parsing Instructions

0xEB extension bit parsing process:

1. Read extension length (2 Byte)
2. Read extension ID (2 Byte)
3. Dispatch to corresponding parser based on extension ID
4. Read data content (length is extension length - 2)

A.2 0xEB Signal Protocol Definition

Bluetooth Fuel Level Protocol (0x0115)

Extension ID: 0x0115

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x0115
4	Manufacturer ID	WORD	0x8115, big-endian
5	Fuel Tank Height	WORD	Big-endian
7	Fuel Percentage	BYTE	0..100%
8	Battery Voltage	WORD	Unit mV, big-endian
10	Temperature	BYTE	int[-127...+127]
11	Acceleration (X,Y,Z) Values	BYTE[3]	Parsed in order, X, Y, Z values respectively

14	Status Flag	BYTE	Bit0: Meter frequency stable status (0 unstable, 1 stable) Bit1: Continuous 5 values calibration fuel tank height status (0 unstable, 1 stable) Bit2: Sensor activation status (0 not activated, 1 activated) Bit3~Bit7: Reserved
15	Device Name Length	BYTE	Maximum 255 Bytes
16	Device Name	String[N]	N Bytes

Changrun Fuel Level Extension Protocol (0x0023)

Extension ID: 0x0023

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Extension ID and data content length, variable length (2+N), maximum 4 channels
2	Extension ID	WORD	0x0023

4	Data	BYTE[N]	6 Bytes per channel, format is percentage, e.g.: 050.12 represents 50.12%

Example Data: 00 08 00 23 30 35 30 2E 31 32 → 050.12 represents 50.12%

Complete Frame Example (Uplink): 7E 02 00 00 27 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 14 55 07 EB 09 00 07 00 23 35 30 2E 31 32 4F 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 10 00 01 02 00 00 1D 7E

Jiutong Fuel Level Extension Protocol (0x004B)

Extension ID: 0x004B

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Extension ID and data content length, variable length (2+N), maximum 4 channels
2	Extension ID	WORD	0x004B
4	Data	BYTE[N]	15 Bytes per channel: A(5Byte) + B(4Byte) + C(6Byte)

Data Sub-Unit Description:

- **A (5 Byte):** Current liquid level height permille, maximum value is 10000
- **B (4 Byte):** Current adjusted fuel level value, 0-4095
- **C (6 Byte):** Divided by 100 is current fuel volume in liters, e.g. 384.09L

Example Data: 0011004b303634343333231383033338343039

Complete Frame Example (Uplink): 7E 02 00 00 27 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 14 55 47 EB 09 00 07 00 4B 36 44 30 00 00 0D 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 11 00 01 02 00 00 1C 7E

Temperature Extension Protocol (0x0003)

Extension ID: 0x0003

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Extension ID and data content length, variable length (2+N), maximum 16 channels
2	Extension ID	WORD	0x0003
4	Data	BYTE[N]	2 Bytes per channel, consisting of integer (bit7: 1 indicates negative) and decimal (unit 0.1°C), initial value 0xFF

Example Data: 000A0003 01C9 0B00 9000 FF008000

- Channel 1: 0x0000 = 0°C
- Channel 2: 0x01C9 = 11.6°C
- Channel 3: 0x9005 = -16.5°C
- Channel 4: 0xFF00 = No temperature probe

Complete Frame Example (Uplink): 7E 02 00 00 28 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 14 56 54 EB 0A 00 08 00 03 00 FA 00 FF 01 22 32 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 12 00 01 02 00 00 1F 7E

Temperature Extension Protocol (EZ427 Dedicated Version) (0x0003)

Extension ID: 0x0003

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Extension ID and data content length, variable length (2+N), maximum 16 channels
2	Extension ID	WORD	0x0003
4	Data	BYTE[N]	2 Bytes per channel (unit 0.1°C), int16_t two's complement represents -3276.8~3276.8°C, 0x8000 indicates invalid or error

Example Data: 8000 024A FF5B 00008000

- Channel 1: 0x8000 = Invalid or error
- Channel 2: 0x024A = 58.6°C
- Channel 3: 0xFF5B = -16.5°C
- Channel 4: 0x0000 = 0°C

Complete Frame Example (Uplink): 7E 02 00 00 32 12 34 56 78 90 12 00 01 00 00 00 00 00 00 00 01 01 65 EC 15 06 BE 20 80 00 00 02 58 00 5A 26 05 25 14 14 54 EB 14 00 12 00 03 16 E4 F9 8E 00 00 00 00 02 26 02 94 03 02 22 B0 27 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 06 00 01 02 00 00 0B 7E

Humidity Extension Protocol (0x00F3)

Extension ID: 0x00F3

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Extension ID and data content length, variable length (2+N), maximum 16 channels
2	Extension ID	WORD	0x00F3
4	Data	BYTE[N]	2 Bytes per channel, unit 0.01%, data 0xFF00 indicates this channel humidity is offline

Example Data: 000A00F3 1730 1740 1750 1760

- 0x1730 = 59.36%
- 0x1740 = 59.52%
- 0x1750 = 59.68%
- 0x1760 = 59.84%

Complete Frame Example (Uplink): 7E 02 00 00 28 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 14 58 54 EB 0A 00 08 00 F3 17 30 17 70 22 C4 4C 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 13 00 01 02 00 00 1E 7E

Video Status (0x00F8)

Extension ID: 0x00F8

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	0006
2	Extension ID	WORD	0x00F8
4	Video Status	DWORD	Bit[31~0]

Bit Definition:

Bit	Field	Description
bit8	Safety Lock Open Status	—
bit9	Codec Heartbeat Exception Status	—
bit30	Intercom Request Flag	—
bit31	4G Module Control Right Status	—
Other bits	Reserved	Default fill 0

HEX Example:00 06 00 F8 80 00 00 00 → Extension length 0006, extension ID 00F8, video status = 0x80000000, bit31=1 (4G module control right status set), all other bits are 0

Complete Frame Example (Uplink): 7E 02 00 00 26 12 34 56 78 90 12 00 01 00 00 00 00 00 00 00 01 01 65 EC 15 06 BE 20 80 00 00 02 58 00 5A 26 05 25 14 44 26 EB 08 00 06 00 F8 C0 00 03 00 85 7E

Load Extension Protocol (0x00E3)

Extension ID: 0x00E3

Field Definition:

Offset	Field	Data Type	Description

0	Extension Length	WORD	Extension ID and data content length, variable length (2+N), maximum 16 channels
2	Extension ID	WORD	0x00E3
4	Data	BYTE[N]	2 Bytes per channel

Example Data: 000600E3 04D2 0929

- Channel 1 AD value: 0x04D2 = 2345
- Channel 2 AD value: 0x0929 = 2345

Complete Frame Example (Uplink): 7E 02 00 00 1C 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 15 01 30 F8 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 15 00 01 02 00 00 18 7E

GSensor Three-Axis Value (0x00EA)

Extension ID: 0x00EA

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	0006
2	Extension ID	WORD	0x00EA
4	Work Status	BYTE	0x01: Normal 0x00: Abnormal
5	GSensor Three-Axis Raw Value	BYTE[3]	x, y, z data

HEX Example:00 06 00 EA 01 1C FE 54 → Extension length 0006, extension ID 00EA, work status = 0x01 (normal), gsensor three-axis raw value = 1C FE 54 (x=0x1C, y=0xFE, z=0x54)

Complete Frame Example (Uplink): 7E 02 00 00 26 12 34 56 78 90 12 00 01 00 00 00 00 00 00 00 01 01 65 EC 15 06 BE 20 80 00 00 02 58 00 5A 26 05 25 14 40 01 EB 08 00 06 00 EA 01 1C FE 54 C0 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 02 00 01 02 00 00 0F 7E

Base Station Upload Protocol (0x00D8)

Extension ID: 0x00D8

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Extension ID and data content length, variable length
2	Extension ID	WORD	0x00D8
4	Data	BYTE[N]	9 Bytes: Country Code (2B) + Operator Code (1B) + Area Code (2B) + Tower Number (4B), all in HEX

Field Breakdown:

- **Country Code:** 2-byte HEX
- **Operator Code:** 1-byte HEX
- **Area Code:** 2-byte HEX
- **Tower Number:** 4-byte HEX

Example Data: 00 0B 00 D8 01 CC 00 26 2C 0A 7F 3E 02

Complete Frame Example (Uplink): 7E 02 00 00 2B 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 15 18 18 EB 0D 00 0B 00 D8 01 CC 01 00 26 0A 7F 3E 02 68 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 16 00 01 02 00 00 1B 7E

WiFi Information Upload (0x013F)

Extension ID: 0x013F

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x013F
4	WiFi MAC Address List Count	BYTE	Maximum 10, starting from 1
5	Data Content	BYTE[N]	See Appendix 1.1

Appendix 1.1 WiFi Data Format

Offset	Field	Data Type	Description
0	WiFi MAC ID	BYTE[6]	Fixed 6 Bytes
6	WiFi Name Length	BYTE	WiFi name may not be transmitted, maximum length 32 Bytes
7	WiFi Name	String[N]	String type

Complete Frame Example (Uplink): 7E 02 00 00 2C 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 15 19 42 EB 0E 00 0C 01 3F 01 AA BB CC DD EE FF E5 AE B6 98 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 17 00 01 02 00 00 1A 7E

Electronic Fence ID and Attributes (0x0123)

Extension ID: 0x0123

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x0123
4	Extension ID Count	BYTE	Maximum 5, starting from 1
5	Data Content	BYTE[N]	See Appendix 2.1

Appendix 2.1 Electronic Fence Data Format

Offset	Field	Data Type	Description
0	ID	DWORD	Electronic fence ID
4	Attributes	WORD	Electronic fence attributes

Complete Frame Example (Uplink): 7E 02 00 00 28 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 15 20 08 EB 0A 00 08 01 23 00 00 30 39 00 01 16 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 18 00 01 02 00 00 15 7E

GPS Satellite Count / Signal-to-Noise Ratio (0x0124)

Extension ID: 0x0124

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x0124

4	Visible Satellite Count	BYTE	—
5	Positioning Satellite Count	BYTE	—
6	SNR Data	BYTE[N]	Maximum 6 Bytes

Complete Frame Example (Uplink): 7E 02 00 00 2A 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 15 21 18 EB 0C 00 0A 01 24 0F 0A 00 01 02 03 04 05 0A 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 19 00 01 02 00 00 14 7E

External Voltage (Power Supply Voltage) (0x00CE)

Extension ID: 0x00CE

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	0004
2	Extension ID	WORD	0x00CE
4	Data Description	WORD	2-byte AD voltage value

Complete Frame Example (Uplink): 7E 02 00 00 24 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 15 22 52 EB 06 00 04 00 CE 2E E0 68 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 1A 00 01 02 00 00 17 7E

MCU Temperature (0x002A)

Extension ID: 0x002A

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	0006
2	Extension ID	WORD	0x002A
4	MCU Temperature 1	WORD	Temperature value, unit 0.1°C
6	MCU Temperature 2	WORD	Temperature value, unit 0.1°C

HEX Example:00 06 00 2A 03 04 00 00 → Extension length 0006, extension ID 002A, MCU temperature 1 = 0x0304 = 772 (77.2°C), MCU temperature 2 = 0x0000 = 0 (0.0°C)

Complete Frame Example (Uplink): 7E 02 00 00 26 12 34 56 78 90 12 00 01 00 00 00 00 00 00 00 01 01 65 EC 15 06 BE 20 80 00 00 02 58 00 5A 26 05 25 13 53 12 EB 08 00 06 00 2A 03 79 00 00 CA 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 02 00 01 02 00 00 0F 7E

Internal Voltage (Battery Voltage) (0x002D)

Extension ID: 0x002D

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	0004
2	Extension ID	WORD	0x002D
4	Data Description	WORD	2 Bytes, unit mV

Example: C54 represents voltage value 3156 mV

Complete Frame Example (Uplink): 7E 02 00 00 24 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 15 23 21 EB 06 00 04 00 2D 0C 54 6F 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 1B 00 01 02 00 00 16 7E

External Voltage (ADC Acquisition Voltage) (0x00E2)

Extension ID: 0x00E2

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	0006, variable length
2	Extension ID	WORD	0x00E2
4	Channel 1	DWORD	4-byte AD voltage value, e.g.: 0x000030D5 represents 12.501V
8	Channel 2	DWORD	4-byte AD voltage value
...	Reserved...	—	Reserved

Complete Frame Example (Uplink): 7E 02 00 00 26 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 15 24 05 EB 08 00 06 00 E2 30 D5 40 74 04 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 1C 00 01 02 00 00 11 7E

ICCID Number (0x00B2)

Extension ID: 0x00B2

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	0x000C
2	Extension ID	WORD	0x00B2

4	Data Description	BYTE[10]	10-byte SIM card ICCID number, in HEX

Example: "89860018190839008096" represents **0x89 0x86 0x00 0x18 0x08 0x39 0x00 0x80 0x96**

Complete Frame Example (Uplink): 7E 02 00 00 36 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 15 25 23 EB 18 00 16 00 B2 38 39 38 36 30 30 31 38 31 39 30 38 33 39 30 30 38 30 39 36 B9 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 1D 00 01 02 00 00 10 7E

Extended Alarm Status Bits (0x0089)

Extension ID: 0x0089

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	0006
2	Extension ID	WORD	0x0089
4	Data Description	DWORD	State[31~0]

Bit Definition:

Bit	Field	Description
bit0	batterySwitch	1: Battery switch-off; 0: Battery switch-on
bit1	Sleep Status	1: Terminal normal status; 0: Terminal sleep status
bit2	Card Swipe Status	1: Normal; 0: Driving without card swipe

bit3	Overspeed Status	1: Normal; 0: Overspeed during time period
bit4	collideAlarm	1: Normal; 0: Collision alarm
bit5	Evaluator Status	1: Evaluator connected normal; 0: Evaluator connection abnormal
bit6	Top Cover Status	1: Top cover closed; 0: Top cover not closed
bit7	Compartment Status	1: Compartment not lifted; 0: Compartment lifted
bit8	accelerateAlarm	1: Normal; 0: Harsh acceleration alarm
bit9	slowDownAlarm	1: Normal; 0: Harsh braking alarm
bit10	Acceleration Sensor Status	1: Acceleration sensor normal; 0: Acceleration sensor abnormal
bit11	illegalMove	1: Normal; 0: Illegal displacement alarm
bit12	Removal Status	1: Normal; 0: Illegal removal
bit13	High Temperature Alarm	1: Normal; 0: High temperature alarm
bit14	Low Temperature Alarm	1: Normal; 0: Low temperature alarm
bit15	shockAlarm	1: Normal; 0: Vibration alarm
bit16	Cut Alarm	1: Normal; 0: Cut alarm (ultrasonic fuel sensor probe)

bit17	Power Loss Alarm	1: Normal; 0: Power loss alarm (cleared upon receiving response)
bit18	Idle Alarm	1: Normal; 0: Idle alarm (cleared upon receiving response)
bit19	LED Display Screen Status	1: LED display screen connected normal; 0: LED display screen fault or not connected
bit20	Taximeter Status	1: Taximeter connected normal; 0: Taximeter fault or not connected
bit21	Multifunction Screen Status	1: Multifunction screen connected normal; 0: Multifunction screen fault or not connected
bit22	Material Shortage Alarm	1: Normal; 0: Material shortage alarm
bit23	Video Connection Status	1: Normal; 0: Abnormal
bit24	ACC Status	1: ACC normal; 0: ACC disconnection alarm
bit25	swerveAlarm	1: Normal; 0: Harsh steering alarm
bit26	ACC Switch Alarm	1: Normal; 0: ACC transition from off to on alarm
bit27	Main Power Switch Status	1: Main power switch normal; 0: Main power switch abnormal
bit28	Burner Switch Status	1: Burner switch normal; 0: Burner switch abnormal

bit29	Vibration Alarm Switch Status	1: Vibration alarm switch off; 0: Vibration alarm switch on
bit30	pseudoLbsState	1: Normal; 0: Pseudo-base station detected
bit31	pseudoLbs	Pseudo-base station alarm

Complete Frame Example (Uplink): 7E 02 00 00 1C 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 15 27 21 CF 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 1E 00 01 02 00 00 13 7E

Extended Alarm Data Bits 2 (0x00C5)

Extension ID: 0x00C5

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	0006
2	Extension ID	WORD	0x00C5
4	Data Description	DWORD	State[0~31], default 0xFFFFFFFF

Bit Definition:

Bit	Field	Description
Bit0	softSwitch	1: Software switch-off; 0: Software switch-on
Bit1	wayBillModel	1: Waybill mode-off; 0: Waybill mode-on
Bit2	isCharge	1: Not charging; 0: Charging

Bit3~Bit4	posStateOne	Positioning status Bit3=0 Bit4=0: Not positioned (neither GPS nor WIFI positioning) Bit3=1 Bit4=0: GPS positioning Bit3=0 Bit4=1: WIFI positioning
Bit5	Unloading Status	1: Not unloaded; 0: Unloaded
Bit6	shockAlarm	1: Normal; 0: Vibration alarm
Bit7	collideAlarm	1: Normal; 0: Collision alarm
Bit8	Arming Status	1: Disarmed; 0: Armed
Bit9	Motion Status	1: Stationary; 0: Moving
Bit10~Bit11	Work Mode	Bit10=1 Bit11=1: Normal work mode Bit10=0 Bit11=1: Deep power saving mode Bit10=1 Bit11=0: Smart power saving mode
Bit12	Force GPS On	1: Not force GPS on; 0: Force GPS on
Bit13	Wired/Wireless Mode	1: Wired mode; 0: Wireless mode
Bit14	Light Sensor Status	0: Light detected; 1: No light detected
Bit15	G-Sensor Mode	1: G-Sensor mode; 0: Non G-Sensor mode
Bit16	Infrared Status	1: Infrared normal; 0: Infrared abnormal

Bit17	Pressure Status	1: Pressure normal; 0: High pressure alarm
Bit18	Tachograph Status	1: Tachograph normal; 0: Tachograph abnormal
Bit19~Bit20	Positioning Mode	Bit19=0 Bit20=1: Single BeiDou positioning Bit19=1 Bit20=0: Single GPS positioning Bit19=0 Bit20=0: Dual-mode positioning
Bit21	Fuel Circuit Function	0: Overspeed warning; 1: Fuel circuit function
Bit22	Fuel Circuit Status	0: Fuel circuit disconnected; 1: Fuel circuit normal
Bit23	Lane Restriction Alarm	0: Lane restriction alarm; 1: Normal
Bit24	Undervoltage Protection Status	0: Undervoltage protection status; 1: Normal status
Bit25	Low Power Mode	0: Low power mode; 1: Non low power mode
Bit26	Customer/ Mode	0: Customer mode; 1: mode
Bit27	Left Turn Overspeed Alarm	0: Left turn overspeed alarm; 1: Normal
Bit28	Right Turn Overspeed Alarm	0: Right turn overspeed alarm; 1: Normal
Bit29	Infrared 1 Trigger	0: Infrared 1 triggered; 1: Infrared 1 not triggered
Bit30	Infrared 2 Trigger	0: Infrared 2 triggered; 1: Infrared 2 not triggered

Bit31	Infrared 3 Trigger	0: Infrared 3 triggered; 1: Infrared 3 not triggered

Complete Frame Example (Uplink): 7E 02 00 00 1C 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 15 28 00 E1 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 1F 00 01 02 00 00 12 7E

Alarm Event Upload (0x012D)

Extension ID: 0x012D

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	0003
2	Extension ID	WORD	0x012D
4	Alarm ID	BYTE	Decimal representation

Alarm ID List:

ID	Name
1	ACC ON
2	ACC OFF
3	Power Restored
4	Power Cut
5	SOS Triggered
6	SOS Cleared
7	Fuel Waste Start (Idle start)
8	Fuel Waste End (Idle end)

9	Positioning Start
10	Positioning End
11	Storage Space Normal
12	SD1 Storage Space Insufficient
13	SD2 Storage Space Insufficient
14	HDD Storage Space Insufficient
15	Local Operator Switched to Other Operator (Dedicated version HW120)
16	Other Operator Switched to Local Operator (Dedicated version HW120)
17	Enter Fence Alarm (Dedicated version HW120)
18	Exit Fence Alarm (Dedicated version HW120)
19	Overspeed Alarm Start (Dedicated version HW120)
20	Overspeed Alarm End (Dedicated version HW120)
21	Timed Report Event (Dedicated version HW120)
22	Distance-based Report Event (Dedicated version HW120)
23	Turning Point Report Event (Dedicated version HW120)
24	Work Time Start (Dedicated version HW120)
25	Work Time End (Dedicated version HW120)
26	input1 Connection Event (Dedicated version HW120)

27	input1 Removal Event (Dedicated version HW120)
28	input2 Connection Event (Dedicated version HW120)
29	input2 Removal Event (Dedicated version HW120)
30	ibutton Login (Dedicated version HW120)
31	ibutton Logout (Dedicated version HW120)
32	Illegal Driving Start (Dedicated version HW120)
33	Illegal Driving End (Dedicated version HW120)
34	Motion Status (Dedicated version HW120)
35	Stationary Status (Dedicated version HW120)
36	GSM Jamming Alarm Start
37	GSM Jamming Alarm End
38	GPS Jamming Alarm Start
39	GPS Jamming Alarm End

Alarm Parameter Report (Customer Customized) (0x012F)

Extension ID: 0x012F

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x012F

4	Alarm List Count	BYTE	Can be triggered simultaneously, indicates count of Appendix 3.1-3.4, starting from 1
5	Alarm List	BYTE[N]	—

Appendix 3.1 Overspeed Alarm List

Offset	Field	Data Type	Description
0	List Length	BYTE	Variable length
1	List Type ID	BYTE	Alarm type 01, decimal
2	Overspeed Value	WORD	Current device speed, unit km/h
4	Overspeed Threshold	WORD	Unit km/h
6	Overspeed Duration	WORD	Unit seconds

Appendix 3.2 Harsh Acceleration Alarm List

Offset	Field	Data Type	Description
0	List Length	BYTE	Variable length
1	List Type ID	BYTE	Alarm type 02, decimal
2	Harsh Acceleration Value	WORD	Device speed at harsh acceleration trigger, km/h/s
4	Harsh Acceleration Threshold	WORD	Unit km/h/s

Appendix 3.3 Harsh Braking Alarm List

Offset	Field	Data Type	Description
0	List Length	BYTE	Variable length
1	List Type ID	BYTE	Alarm type 03, decimal
2	Harsh Braking Value	WORD	Device speed at harsh braking trigger, km/h/s
4	Harsh Braking Threshold	WORD	Unit km/h/s

Appendix 3.4 Harsh Steering Alarm List

Offset	Field	Data Type	Description
0	List Length	BYTE	Variable length
1	List Type ID	BYTE	Alarm type 04, decimal
2	Harsh Steering Angular Velocity	WORD	Device gravity at harsh steering trigger, unit g's
4	Harsh Steering Angular Velocity Threshold	WORD	Harsh steering threshold gravity unit g's

---

Status Upload (0x0137)

Extension ID: 0x0137

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length

2	Extension ID	WORD	0x0137
4	Status ID 1	DWORD	In HEX, 1 represents valid, 0 represents invalid
...	...	...	...
N	Status ID N	DWORD	Extensible

Status ID 1 Bit Definition:

Bit	Field	Description
bit0~bit1	Network Type	00: Unknown; 01: 2G network; 02: 4G network; 03: 5G network
bit2~bit31	Reserved	Fill with 0 by default

Complete Frame Example (Uplink): 7E 02 00 00 3D 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 16 06 21 EB 1F 00 1D 01 2F 04 06 01 00 50 00 3C 00 3C 05 02 00 1E 00 32 05 03 00 1E 00 32 05 04 00 0F 00 14 43 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 22 00 01 02 00 00 2F 7E

IMEI (0x00D5)

Extension ID: 0x00D5

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x00D5
4	Data Description	BYTE[15]	15 Bytes ASCII code

Example: 383635373330303736313136363836 = "865730076116686"

Complete Frame Example (Uplink): 7E 02 00 00 31 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 16 07 25 EB 13 00 11 00 D5 31 32 33 34 35 36 37 38 39 30 31 32 33 34 35 C9 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 23 00 01 02 00 00 2E 7E

---

ibutton Extension Protocol (0x0120)

Extension ID: 0x0120

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Extension ID and data content length, fixed length
2	Extension ID	WORD	0x0120
4	Data	BYTE[10]	Login/logout status Byte0: 0x00 logout, 0x01 login, 0x02 (no login status), 0xFF unknown error Authorization status Byte1: 0x00 not authorized; 0x01 authorized Button data Byte2-byte9: One 64-bit ID, i.e., 8 Bytes

Example Data: 01E32C9F01000044

Complete Frame Example (Uplink): 7E 02 00 00 2C 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 16 08 44 EB 0E 00 0C 01 20 00 00 01 E3 2C 9F 01 00 00 44 6A 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 24 00 01 02 00 00 29 7E

Driver Information (0x0126)

Extension ID: 0x0126

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x0126
4	Status	BYTE	0x01: Driver card login; 0x02: Driver card logout
5	Driver Name Length	BYTE	—
6	Driver Name	String[N]	N Bytes
7+N	Business License Number Length	BYTE	—
8+N	Business License Number	String[M]	M Bytes

Complete Frame Example (Uplink): 7E 02 00 00 33 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 16 09 41 EB 15 00 13 01 26 01 BB C6 C7 EF B7 E3 00 31 32 33 34 36 37 38 39 30 53 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 25 00 01 02 00 00 28 7E

Device Historical Speed (0x0127)

Extension ID: 0x0127

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x0127
4	Speed Count	BYTE	Maximum 60, starting from 1, each speed occupies 1 byte
5	Historical Speed Data	BYTE[N]	Unit km/h, when speed is 255, indicates invalid value

Note: When upload interval is set to 30 seconds, at upload time, 29 historical speeds are uploaded.

Complete Frame Example (Uplink): 7E 02 00 00 28 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 16 10 30 EB 0A 00 08 01 27 03 3C 46 50 00 00 38 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 26 00 01 02 00 00 2B 7E

Storage Status (0x012A)

Extension ID: 0x012A

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x012A
5	List Count	BYTE	Indicates Appendix 4.1 media count

6	Storage Media Data Content	BYTE[N]	Can support multiple SD1 status, SD2 status, hard drive status

Appendix 4.1 Storage Status Data Format

Offset	Field	Data Type	Description
0	List Data Length	BYTE	—
1	Storage Type	BYTE	0-SD1; 1-SD2; 2-Hard drive; 3-EMMC/Flash
2	Manufacturer Information Length	BYTE	Variable length
3+N	Manufacturer Information	String[N]	—
4+N	Status	BYTE	0-Normal; 1-Does not exist; 2-Not formatted; 3-Disk damaged; 4-Disk abnormal; 0xFF-Unknown error
5+N	Total Capacity	DWORD	Unit M
9+N	Overwrite Count	DWORD	Unit M
13+N	Format Time	BCD[6]	Year/month/day/hour/minute/second
19+N	Storage Media CID Information Length	BYTE	—
20+N	Storage Media CID Information	String	—

Complete Frame Example (Uplink): 7E 02 00 00 44 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 16 30 30 EB 26 00 24

01 2A 01 20 00 06 41 42 43 31 32 33 03 00 00 7D 01 00 00 00 03 E8 26 05 12 16 30 30 08 43 49 44 31 32 33 34 35 C2 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 27 00 01 02 00 00 2A 7E

Device Timezone Status Upload (0x0131)

Extension ID: 0x0131

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x0131
3	Timezone Information	BYTE	1-byte timezone, -11 to 12, positive number indicates east timezone, negative number indicates west timezone
4	Half Timezone Information	BYTE	0, 15, 30, 45
5	Timezone Effective Flag	BYTE	0: Timezone effective to system; 1: System fixed to 0 timezone, timezone only effective to watermark
6~10	Reserved	BYTE[5]	Reserved

Complete Frame Example (Uplink): 7E 02 00 00 24 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 16 31 03 EB 06 00 04 01 31 08 1E 0F 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 28 00 01 02 00 00 25 7E

Device Last Network Exception Status Upload (0x0134)

Extension ID: 0x0134

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x0134
3	Basic Network Exception Reason	WORD	Bit0: SIM card (CPIN status) 1 normal 0 abnormal Bit1: CSQ signal weak 1 normal 0 abnormal Bit2: Operator info abnormal (AT+COPS?) 1 normal 0 abnormal Bit3: CREG 1 normal 0 abnormal Bit4: CGREG 1 normal 0 abnormal Bit5: PDP 1 normal 0 abnormal Bit6: Module initialization 1 normal 0 abnormal Bit6-Bit15 reserved

5	Link 1 Service Network Exception Reason	BYTE	By IP link, uploaded when effective Bit0: Registered platform no response 1 normal 0 abnormal Bit1: Authentication platform no response 1 normal 0 abnormal Bit2: Location data platform no response 1 normal 0 abnormal Bit3: Platform link actively disconnected 1 normal 0 abnormal Bit4-Bit7 reserved
6	Link 2 Service Network Exception Reason	BYTE	By IP link, uploaded when effective Bit0: Registered platform no response 1 normal 0 abnormal Bit1: Authentication platform no response 1 normal 0 abnormal Bit2: Location data platform no response 1 normal 0 abnormal Bit3: Platform link actively disconnected 1 normal 0 abnormal Bit4-Bit7 reserved

7	Link 3 Service Network Exception Reason	BYTE	By IP link, uploaded when effective Bit0: Registered platform no response 1 normal 0 abnormal Bit1: Authentication platform no response 1 normal 0 abnormal Bit2: Location data platform no response 1 normal 0 abnormal Bit3: Platform link actively disconnected 1 normal 0 abnormal Bit4-Bit7 reserved
N	IP Link Extensible	—	—

Complete Frame Example (Uplink): 7E 02 00 00 23 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 16 31 27 EB 05 00 03 01 34 07 3C 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 29 00 01 02 00 00 24 7E

Driver Login Information (0x0135)

Extension ID: 0x0135

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x0135

4	Status	BYTE	0x00: Invalid; 0x01: Driver card login; 0x02: Driver card logout
5	IC Card Content Length	BYTE	—
6	IC Card Content	String[N]	N Bytes

Complete Frame Example (Uplink): 7E 02 00 00 23 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 15 53 50 EB 05 00 03 01 37 02 2C 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 21 00 01 02 00 00 2C 7E

---

gsensor Vibration Threshold (0x0136)

Extension ID: 0x0136

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x0136
4	Vector Value 1	WORD	Unit mg
...	......	......	......
4+N*2	Vector Value N	WORD	Unit mg, N maximum

Complete Frame Example (Uplink): 7E 02 00 00 26 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 16 32 32 EB 08 00 06 01 36 00 64 00 96 D0 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 2B 00 01 02 00 00 26 7E

Device Version Information Upload (0x0140)

Extension ID: 0x0140

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length, maximum version data length 40 Bytes
2	Extension ID	WORD	0x0140
4	MCU Version Information	String[N]	String type
5+N	DVR Version Information	String[M]	String type

Bluetooth Fuel Level Protocol (0x0121)

Extension ID: 0x0121

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x0121, this extension field does not exist when MAC ID is not configured

4	Channel ID	BYTE	Maximum 4, starting from 1, indicates Appendix 5.1 data count

Appendix 5.1 Bluetooth Fuel Level Data Format

Offset	Field	Data Type	Description
0	List Length	BYTE	Variable length
1	MAC ID Length	BYTE	Maximum 20, variable length
2	MAC ID	String[N]	N Bytes
3+N	Manufacturer ID	WORD	Big-endian, 0x8115 represents 0x8115
5+N	Fuel Tank Liquid Level Height	WORD	Big-endian
7+N	Fuel Percentage	BYTE	0..100%
8+N	Battery Voltage	WORD	Unit mV
10+N	Temperature	WORD	Unit 0.1°C, int16_t two's complement represents -3276.8~3276.8°C, 0x8000 indicates invalid or error
11+N	Status	BYTE	Work status Bit0: 1 normal 0 abnormal; Bit1-Bit7: Reserved

Complete Frame Example (Uplink): 7E 02 00 00 2C 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 17 19 25 EB 0E 00 0C 01 25 00 32 00 00 13 88 00 00 27 10 94 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 42 00 01 02 00 00 4F 7E

Bluetooth Temperature/Humidity Sensor (0x0122)

Extension ID: 0x0122

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x0122, this extension field does not exist when MAC ID is not configured
4	Channel ID	BYTE	Maximum 8 channels, starting from 1, indicates Appendix 6.1 data count

Appendix 6.1 Temperature/Humidity Type A Data Format

Offset	Field	Data Type	Description
0	List Length	BYTE	Variable length
1	MAC ID Length	BYTE	Maximum 20 Bytes
2+N	MAC ID	String[N]	—
3+N	Manufacturer ID	WORD	0x8115
5+N	Temperature	WORD	Unit 0.1°C, int16_t two's complement represents -3276.8~3276.8°C, 0x8000 indicates invalid or error

7+N	Humidity	BYTE	0..100%
8+N	Battery Voltage	WORD	Unit mV
10+N	Angle	BYTE[3]	Angle data
14+N	Motion Status	WORD	Motion status and count
16+N	Flag	BYTE	bit0: Temperature value exists; bit1: Humidity value exists; bit2: Magnetic sensor exists; bit3: Magnetic sensor status; bit4: Motion sensor counter; bit5: Motion sensor angle; bit6: Low battery indicator; bit7: Battery voltage value exists

Complete Frame Example (Uplink): 7E 02 00 00 26 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 16 43 39 EB 08 00 06 01 29 01 2C 01 90 FB 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 31 00 01 02 00 00 3C 7E

AIBOX Fuel Gauge/Mileage Upload (0x0125)

Extension ID: 0x0125

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x0125

4	Fuel Gauge Data Type	BYTE	0-Percentage; 1-Fuel gauge value
5	Fuel Percentage	BYTE	0..100%
6	Fuel Gauge Value	DWORD	Unit 0.1L
10	Mileage	DWORD	Unit 0.1km

Device Internal Temperature Upload (0x0129)

Extension ID: 0x0129

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x0129
4	Temperature Data	BYTE[N]	2 Bytes per temperature type, unit 0.1°C, int16_t two's complement represents -3276.8~3276.8°C, 0x8000 indicates invalid or error

Note: byte0-byte1: temp1st MCU internal temperature or DVR internal temperature; byte2-byte5: temp2nd reserved. When device has both MCU and SOC internal temperatures, byte0-byte1 represents MCU temperature, byte2-byte3 represents SOC temperature.

Complete Frame Example (Uplink): 7E 02 00 00 29 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 16 44 13 EB 0B 00 09 01 2B 01 09 C4 01 13 88 32 0F 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 32 00 01 02 00 00 3F 7E

Fuel Level Data Upload (0x012B)

Extension ID: 0x012B

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x012B
5	List Count	BYTE	Indicates Appendix 7.1 extended fuel level count
6	Fuel Level Data Content	BYTE[N]	Can support multiple fuel level data

Appendix 7.1 Fuel Level Data Format

Complete Frame Example (Uplink): 7E 02 00 00 28 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 16 44 45 EB 0A 00 08 01 2C 12 34 56 78 01 5E 6C 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 33 00 01 02 00 00 3E 7E

Offset	Field	Data Type	Description
0	List Data Length	BYTE	—
1	Channel ID	BYTE	Range 0-0xFF
2	Temperature	WORD	Unit 0.1°C, int16_t two's complement represents -3276.8~3276.8°C, 0x8000 indicates invalid or error

4	Fuel Tank Liquid Level Unit	BYTE	0-Unknown; 1-Liter; 2-Gallon; 3-Fuel tank height
5	Fuel Amount in Tank	DWORD	0 is analog (unit unknown); 1 or 2 represents fuel amount, precision 0.1, unit is liter or gallon; 3 represents fuel tank height, unit 0.1mm
9	Fuel Percentage	BYTE	0-100%
10	Status	BYTE	Work status Bit0: 1 normal 0 abnormal; Bit1-Bit7: Reserved

One-Wire ID and Temperature Extension Protocol (0x012C)

Extension ID: 0x012C

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x012C
4	Channel ID	BYTE	Maximum 8 channels, starting from 1, indicates Appendix 8.1 data count

Appendix 8.1 One-Wire Data Format

Offset	Field	Data Type	Description

0	List Length	BYTE	Variable length
1	ID Length	BYTE	Maximum 20 Bytes
2	ID Data	String[N]	In HEX representation
3+N	Temperature	WORD	Unit 0.1°C, int16_t two's complement represents -3276.8~3276.8°C, 0x8000 indicates invalid or error

Complete Frame Example (Uplink): 7E 02 00 00 37 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 17 26 25 EB 19 00 17 01 2E 14 00 00 8C A0 00 00 38 40 00 00 00 05 00 00 01 2C 00 00 00 0A 4B 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 43 00 01 02 00 00 4E 7E

Device Statistics Data Upload (0x012E)

Extension ID: 0x012E

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x012E
4	Card Fatigue Time	BYTE[N]	Indicates Appendix 9.1 data
...	Reserved Additional Data	...	...

Appendix 9.1 Statistics Data Format

Offset	Field	Data Type	Description
0	List Length	BYTE	Variable length
1	Current Driver Card Daily Cumulative Time	DWORD	Unit seconds, default triggers cumulative fatigue driving when exceeding 10 hours
5	Current Driver Card Single Trip Time	DWORD	Unit seconds, default triggers fatigue driving when exceeding 4 hours
10	Vehicle Parking Count	DWORD	Calculated by day, reset at midnight every day
14	Parking Duration	DWORD	Single parking duration, reset each time vehicle starts
18	Overspeed Count	DWORD	Calculated by day, reset at midnight every day

---

Bluetooth Door Sensor (0x0130)

Extension ID: 0x0130

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length

2	Extension ID	WORD	0x0130, this extension field does not exist when MAC ID is not configured
4	Channel ID	BYTE	Maximum 8 channels, starting from 1, indicates Appendix 10.1 data count

Appendix 10.1 Bluetooth Door Sensor Data Format

Offset	Field	Data Type	Description
0	List Length	BYTE	Variable length
1	Door Sensor Status	BYTE	0 closed; 1 open
2	Battery Voltage	BYTE[3]	byte0 unit: 0-mV; 1-Percentage; byte1-byte2 voltage value
5	Work Status	BYTE	Work status Bit0: 1 normal 0 abnormal; Bit1-Bit7: Reserved
6	Open Count	DWORD	Reserved for use, not as standard function
10	Close Count	DWORD	Reserved for use, not as standard function
14	Removal Alarm Count	DWORD	Reserved for use, not as standard function
15	MAC ID Length	BYTE	Maximum 20 Bytes, reserved for use

16	MAC ID	String[N]	Reserved for use
16+N	Temperature	WORD	Unit 0.1°C, int16_t two's complement represents -3276.8~3276.8°C, 0x8000 indicates invalid or error

Complete Frame Example (Uplink): 7E 02 00 00 2C 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 16 33 21 EB 0E 00 0C 01 40 56 31 2E 30 31 2C 56 32 2E 30 5E 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 2C 00 01 02 00 00 21 7E

Load Sensor Upload Protocol (Overseas Standard) (0x0132)

Extension ID: 0x0132

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x0132
3	Sensor List Count	BYTE	Indicates Appendix 11.1 count, starting from 1, maximum 8
4	Data List	BYTE[N]	—

Appendix 11.1 Load Sensor Data Format

Offset	Field	Data Type	Description
0	List Length	BYTE	Variable length

1	External Device ID	WORD	For RS485 use, fill 0 if other interfaces do not exist
3	Connection Status	BYTE	0 abnormal; 1 normal
4	AD Value	DWORD	—
8	Load Value	DWORD	Optional, unit 0.1 KG
12	Rated Load Value	DWORD	Optional, unit 0.1 KG
16	Overload Threshold	DWORD	Optional, unit 0.1 KG
20	Loading Count	DWORD	Optional
24	Load Status	BYTE	Optional: 01-Empty; 02-Full; 03-Overload; 04-Loading; 05-Unloading; 06-Light load; 07-Heavy load
25	Temperature	WORD	Optional, unit 0.1°C, int16_t two's complement represents -3276.8~3276.8°C, 0x8000 indicates invalid or error

Complete Frame Example (Uplink): 7E 02 00 00 3A 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 16 57 57 EB 1C 00 1A 01 21 01 06 AA BB CC DD EE FF AA BB CC DD EE FF 81 95 03 E8 32 0B B8 00 FA 01 A3 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 3F 00 01 02 00 00 32 7E

Passenger Counter Extension Protocol (0x0133)

Extension ID: 0x0133

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x0133
3	Sensor List Count	BYTE	Indicates Appendix 12.1 count, starting from 1, maximum 8
4	Data List	BYTE[N]	Uploaded when this data segment has changes

Appendix 12.1 Passenger Counter Data Format

Offset	Field	Data Type	Description
0	List Length	BYTE	Variable length
1	External Device ID	WORD	For RS485 use, fill 0 if other interfaces do not exist
3	Connection Status	BYTE	0 abnormal; 1 normal
4	Cumulative Front Door Total Entry	DWORD	—
8	Cumulative Front Door Total Exit	DWORD	—
12	Cumulative Rear Door Total Entry	DWORD	—
16	Cumulative Rear Door Total Exit	DWORD	—

Complete Frame Example (Uplink): 7E 02 00 00 37 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 17 06 41 EB 19 00 17 01 22 01 06 AA BB CC DD EE FF 81 95 00 FA 3C 0B B8 01 02 03 00 0A FF E3 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 41 00 01 02 00 00 4C 7E

Bluetooth Angle Sensor (Dedicated Version) (0x0138)

Extension ID: 0x0138

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x0138, this extension field does not exist when MAC ID is not configured
4	Channel ID	BYTE	Maximum 8 channels, starting from 1, indicates Appendix 13.1 data count

Appendix 13.1 Bluetooth Angle Sensor Data Format

Offset	Field	Data Type	Description
0	List Length	BYTE	Variable length

1	Angle Sensor Mode	BYTE	0x06 Mode 6 (angle sensor mode enabled), range: 0, 4, 5, 6, 9, 10. For mode definitions, refer to DU-BLE data manual and user manual
2	Event Notification/RPM	BYTE	Mode 6: 0x01 event notification enabled - angle out of range
3	Tilt Angle/Total Rotation Counter	WORD	byte1: 0..180 degrees; byte2: 0..65535 rotations
5	Temperature	BYTE	-70...125°C, 0x17 (HEX) = 23°C
6	Roll Angle	WORD	0...180 (0=-90°, 90=0°, 180=90°) 0x0041 (big-endian) = 65°
8	Pitch Angle	WORD	0...360 (0=-180°, 180=0°, 360=180°) 0x0041 (big-endian) = 65°
10	Built-in Battery Voltage	BYTE	2...4 unit 0.1V 0x1D (HEX)=29→2.9V
11	Firmware Version	BYTE	0...255 0x66 (HEX)=102 version
12	MAC ID Length	BYTE	Maximum 20 Bytes
13	MAC ID	String[N]	—

Complete Frame Example (Uplink): 7E 02 00 00 29 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 16 46 54 EB 0B 00 09

01 30 00 0B B8 00 01 00 02 85 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 34 00 01 02 00 00 39 7E

Storage Format Alarm Upload (0x0139)

Extension ID: 0x0139

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x0139
5	List Count	BYTE	Indicates Appendix 14.1 media count
6	Storage Media Data Content	BYTE[N]	Can transmit multiple SD1 status, SD2 status, hard drive status

Appendix 14.1 Storage Format Status Data Format

Offset	Field	Data Type	Description
0	List Data Length	BYTE	—
1	Format Type	BYTE	0 automatic format; 1 manual format
2	Storage Type	BYTE	0-SD; 1-Hard drive; 2-EMMC/Flash
3	Same Type Storage Media Number	BYTE	Used to distinguish multiple storage devices, starting from 1, decimal

4	Format Result	BYTE	0x00-success; 0x01-failure
5	Format Time	BCD[6]	BCD format time

Complete Frame Example (Uplink): 7E 02 00 00 3E 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 17 33 18 EB 20 00 1E 01 32 01 1A 00 01 00 00 00 03 E8 00 00 27 10 00 00 4E 20 00 00 61 A8 00 00 00 0A 04 00 FA B1 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 45 00 01 02 00 00 48 7E

Three-Axis Acceleration Sensor Data Upload (0x013A)

Extension ID: 0x013A

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x013A
5	Three-Axis Data Total Length	BYTE	Length is multiple of 3
6	x,y,z Data	BYTE[N]	Data format arranged by xyz, xyz

Complete Frame Example (Uplink): 7E 02 00 00 2C 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 16 48 17 EB 0E 00 0C 01 33 00 01 00 64 00 50 00 28 00 3C 5F 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 36 00 01 02 00 00 3B 7E

Temperature Sensor Status (Standard Version Protocol) (0x013B)

Extension ID: 0x013B

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x013B
5	List Count	BYTE	Indicates Appendix 15.1 temperature sensor count, maximum 20
6	Sensor Status Data Content	BYTE[N]	See Appendix 15.1

Appendix 15.1 Sensor Data Format

Offset	Field	Data Type	Description
0	List Data Length	BYTE	—
1	Temperature	WORD	Unit 0.01°C, int16_t two's complement represents -327.68~327.67°C
3	Voltage	WORD	Big-endian, 0x0CD6 represents 3030 mV
5	MAC ID	BYTE[6]	In HEX

11	Device Status	BYTE	bit0: Enabled status 1 enabled 0 disabled bit1: Connection status 1 present 0 lost (no data received for 30 minutes) bit2-7: Reserved

Complete Frame Example (Uplink): 7E 02 00 00 36 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 19 36 24 EB 18 00 16 01 38 01 12 04 01 41 0A 17 00 5A 00 B4 1D 66 06 AA BB CC DD EE FF E9 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 46 00 01 02 00 00 4B 7E

Ultrasonic Fuel Level Sensor (Standard Version Protocol) (0x013C)

Extension ID: 0x013C

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x013C
5	List Count	BYTE	Indicates Appendix 16.1 sensor count, maximum 6
6	Sensor Status Data Content	BYTE[N]	See Appendix 16.1

Appendix 16.1 Ultrasonic Fuel Level Data Format

Offset	Field	Data Type	Description
0	List Data Length	BYTE	—

1	Liquid Level Height	WORD	Unit 1mm
3	Fuel Percentage	BYTE	0-100%
4	MAC ID	BYTE[6]	In HEX
10	Device Status	BYTE	bit0: Connection status 1 present 0 lost (no data received for 3 minutes) bit1: Tilt status 1 tilted 0 normal bit2-3: Data validity 00 poor 01 excellent 10 good bit4: Power loss alarm 1 triggered 0 normal bit5: Enabled status 1 enabled 0 disabled bit6-7: Reserved

Complete Frame Example (Uplink): 7E 02 00 00 2A 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 16 51 06 EB 0C 00 0A 01 39 00 00 26 05 12 16 51 30 38 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 39 00 01 02 00 00 34 7E

Door Sensor (Standard Version Protocol) (0x013D)

Extension ID: 0x013D

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length

2	Extension ID	WORD	0x013D
5	List Count	BYTE	Indicates Appendix 17.1 temperature sensor count, maximum 20
6	Sensor Status Data Content	BYTE[N]	See Appendix 17.1

Appendix 17.1 Door Sensor Data Format

Offset	Field	Data Type	Description
0	List Data Length	BYTE	—
1	Device Status	BYTE	bit0: Door sensor status 0 closed 1 open; bit1: Enabled status 1 enabled 0 disabled; bit2: Connection status 1 present 0 lost (no data received for 30 minutes); bit3-7: Reserved
2	Voltage	WORD	Big-endian, 0x0CD6 represents 3030 mV
4	MAC ID	BYTE[6]	In HEX

Complete Frame Example (Uplink): 7E 02 00 00 28 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 16 52 17 EB 0A 00 08 01 3A 00 03 00 09 00 0C 6F 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 3A 00 01 02 00 00 37 7E

RFID Card Reader (Standard Version Protocol) (0x013E)

Extension ID: 0x013E

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x013E
3	Login Status	BYTE	1 login; 0 logout
4	Card Number	String	Maximum 15 bytes

Complete Frame Example (Uplink): 7E 02 00 00 2D 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 16 52 57 EB 0F 00 0D 01 3B 09 C4 0B B8 AA BB CC DD EE FF 03 41 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 3B 00 01 02 00 00 36 7E

Step Count Accumulation (0x0105)

Extension ID: 0x0105

Field Definition:

Offset	Field	Data Type	Description
0	Extension Length	WORD	Variable length
2	Extension ID	WORD	0x0105
4	Step Count	DWORD	Unit 1 step, unsigned integer, 0xFFFFFFFF indicates invalid

Complete Frame Example (Uplink): 7E 02 00 00 26 12 34 56 78 90 12 00 01 00 00 00 00 00 01 00 00 01 57 FB 60 06 BE 20 48 00 64 00 00 00 00 26 05 12 19 38 35 EB 08 00 06 01 05 00 01 86 A0 34 7E			

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 49 00 01 02 00 00 44 7E